Filter Summary Report: CG,TIA,Automated,NO,L,simple,Z1,Z5

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Contents

1 Examined $H(z)$ for CG TIA Automated NO L simple Z1 Z5: $\frac{Z_1 Z_5 g_m - Z_1}{2 Z_1 g_m + 1}$

$$H(z) = \frac{Z_1 Z_5 g_m - Z_1}{2 Z_1 g_m + 1}$$

2 AP

3 BP

4 BS

5 GE

6 HP

7 LP

8 X-INVALID-NUMER

8.1 X-INVALID-NUMER-1 $Z(s) = \left(\frac{1}{C_1 s},\ \infty,\ \infty,\ \infty,\ \frac{R_5}{C_5 R_5 s + 1},\ \infty \right)$

$$H(s) = \frac{- C_5 R_5 s + R_5 g_m - 1}{C_1 C_5 R_5 s^{2} + 2 g_m + s \left(C_1 + 2 C_5 R_5 g_m\right)}$$

Parameters:

Q: $\frac{\sqrt{2} \sqrt{C_1} \sqrt{C_5} \sqrt{R_5} \sqrt{g_m}}{C_1 + 2 C_5 R_5 g_m}$
wo: $\frac{\sqrt{2} \sqrt{g_m}}{\sqrt{C_1} \sqrt{C_5} \sqrt{R_5}}$
bandwidth: $\frac{C_1 + 2 C_5 R_5 g_m}{C_1 C_5 R_5}$
K-LP: $\frac{R_5 g_m - 1}{2 g_m}$
K-HP: 0
K-BP: $- \frac{C_5 R_5}{C_1 + 2 C_5 R_5 g_m}$
Qz: None
Wz: None

8.2 X-INVALID-NUMER-2 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1},\ \infty,\ \infty,\ \infty,\ \frac{R_5}{C_5 R_5 s + 1},\ \infty \right)$

$$H(s) = \frac{- C_5 R_1 R_5 s + R_1 R_5 g_m - R_1}{C_1 C_5 R_1 R_5 s^{2} + 2 R_1 g_m + s \left(C_1 R_1 + 2 C_5 R_1 R_5 g_m + C_5 R_5\right) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_1} \sqrt{C_5} \sqrt{R_1} \sqrt{R_5} \sqrt{2 R_1 g_m + 1}}{C_1 R_1 + 2 C_5 R_1 R_5 g_m + C_5 R_5}$
wo: $\frac{\sqrt{2 R_1 g_m + 1}}{\sqrt{C_1} \sqrt{C_5} \sqrt{R_1} \sqrt{R_5}}$
bandwidth: $\frac{C_1 R_1 + 2 C_5 R_1 R_5 g_m + C_5 R_5}{C_1 C_5 R_1 R_5}$
K-LP: $\frac{R_1 R_5 g_m - R_1}{2 R_1 g_m + 1}$
K-HP: 0
K-BP: $- \frac{C_5 R_1 R_5}{C_1 R_1 + 2 C_5 R_1 R_5 g_m + C_5 R_5}$
Qz: None
Wz: None

9 X-INVALID-ORDER

9.1 X-INVALID-ORDER-1 $Z(s) = (R_1,\ \infty,\ \infty,\ \infty,\ R_5,\ \infty)$

$$H(s) = \frac{R_1 R_5 g_m - R_1}{2R_1 g_m + 1}$$

9.2 X-INVALID-ORDER-2 $Z(s) = \left(R_1,\ \infty,\ \infty,\ \infty,\ \frac{1}{C_5 s},\ \infty\right)$

$$H(s) = \frac{-C_5 R_1 s + R_1 g_m}{s\left(2C_5 R_1 g_m + C_5\right)}$$

9.3 X-INVALID-ORDER-3 $Z(s) = \left(R_1,\ \infty,\ \infty,\ \infty,\ \frac{R_5}{C_5 R_5 s+1},\ \infty\right)$

$$H(s) = \frac{-C_5 R_1 R_5 s + R_1 R_5 g_m - R_1}{2R_1 g_m + s\left(2C_5 R_1 R_5 g_m + C_5 R_5\right) + 1}$$

9.4 X-INVALID-ORDER-4 $Z(s) = \left(R_1,\ \infty,\ \infty,\ \infty,\ R_5 + \frac{1}{C_5 s},\ \infty\right)$

$$H(s) = \frac{R_1 g_m + s\left(C_5 R_1 R_5 g_m - C_5 R_1\right)}{s\left(2C_5 R_1 g_m + C_5\right)}$$

9.5 X-INVALID-ORDER-5 $Z(s) = \left(\frac{1}{C_1 s},\ \infty,\ \infty,\ \infty,\ R_5,\ \infty\right)$

$$H(s) = \frac{R_5 g_m - 1}{C_1 s + 2g_m}$$

9.6 X-INVALID-ORDER-6 $Z(s) = \left(\frac{1}{C_1 s},\ \infty,\ \infty,\ \infty,\ \frac{1}{C_5 s},\ \infty\right)$

$$H(s) = \frac{-C_5 s + g_m}{C_1 C_5 s^2 + 2C_5 g_m s}$$

9.7 X-INVALID-ORDER-7 $Z(s) = \left(\frac{1}{C_1 s},\ \infty,\ \infty,\ \infty,\ R_5 + \frac{1}{C_5 s},\ \infty\right)$

$$H(s) = \frac{g_m + s\left(C_5 R_5 g_m - C_5\right)}{C_1 C_5 s^2 + 2C_5 g_m s}$$

9.8 X-INVALID-ORDER-8 $Z(s) = \left(\frac{R_1}{C_1 R_1 s+1},\ \infty,\ \infty,\ \infty,\ R_5,\ \infty\right)$

$$H(s) = \frac{R_1 R_5 g_m - R_1}{C_1 R_1 s + 2R_1 g_m + 1}$$

9.9 X-INVALID-ORDER-9 $Z(s) = \left(\frac{R_1}{C_1 R_1 s+1},\ \infty,\ \infty,\ \infty,\ \frac{1}{C_5 s},\ \infty\right)$

$$H(s) = \frac{-C_5 R_1 s + R_1 g_m}{C_1 C_5 R_1 s^2 + s\left(2C_5 R_1 g_m + C_5\right)}$$

9.10 X-INVALID-ORDER-10 $Z(s) = \left(\frac{R_1}{C_1 R_1 s+1},\ \infty,\ \infty,\ \infty,\ R_5 + \frac{1}{C_5 s},\ \infty\right)$

$$H(s) = \frac{R_1 g_m + s \left(C_5 R_1 R_5 g_m - C_5 R_1\right)}{C_1 C_5 R_1 s^2 + s \left(2 C_5 R_1 g_m + C_5\right)}$$

9.11 X-INVALID-ORDER-11 $Z(s) = \left(R_1 + \frac{1}{C_1 s},\ \infty,\ \infty,\ \infty,\ R_5,\ \infty\right)$

$$H(s) = \frac{R_5 g_m + s \left(C_1 R_1 R_5 g_m - C_1 R_1\right) - 1}{2 g_m + s \left(2 C_1 R_1 g_m + C_1\right)}$$

9.12 X-INVALID-ORDER-12 $Z(s) = \left(R_1 + \frac{1}{C_1 s},\ \infty,\ \infty,\ \infty,\ \frac{1}{C_5 s},\ \infty\right)$

$$H(s) = \frac{- C_1 C_5 R_1 s^2 + g_m + s \left(C_1 R_1 g_m - C_5\right)}{2 C_5 g_m s + s^2 \left(2 C_1 C_5 R_1 g_m + C_1 C_5\right)}$$

9.13 X-INVALID-ORDER-13 $Z(s) = \left(R_1 + \frac{1}{C_1 s},\ \infty,\ \infty,\ \infty,\ R_5 + \frac{1}{C_5 s},\ \infty\right)$

$$H(s) = \frac{g_m + s^2 \left(C_1 C_5 R_1 R_5 g_m - C_1 C_5 R_1\right) + s \left(C_1 R_1 g_m + C_5 R_5 g_m - C_5\right)}{2 C_5 g_m s + s^2 \left(2 C_1 C_5 R_1 g_m + C_1 C_5\right)}$$

10 X-INVALID-WZ

10.1 X-INVALID-WZ-1 $Z(s) = \left(R_1 + \frac{1}{C_1 s},\ \infty,\ \infty,\ \infty,\ \frac{R_5}{C_5 R_5 s+1},\ \infty\right)$

$$H(s) = \frac{- C_1 C_5 R_1 R_5 s^2 + R_5 g_m + s \left(C_1 R_1 R_5 g_m - C_1 R_1 - C_5 R_5\right) - 1}{2 g_m + s^2 \left(2 C_1 C_5 R_1 R_5 g_m + C_1 C_5 R_5\right) + s \left(2 C_1 R_1 g_m + C_1 + 2 C_5 R_5 g_m\right)}$$

Parameters:

Q: $\frac{2 \sqrt{2} \sqrt{C_1} \sqrt{C_5} R_1 \sqrt{R_5} g_m \sqrt{\frac{g_m}{2 R_1 g_m + 1}} + \sqrt{2} \sqrt{C_1} \sqrt{C_5} \sqrt{R_5} \sqrt{\frac{g_m}{2 R_1 g_m + 1}}}{2 C_1 R_1 g_m + C_1 + 2 C_5 R_5 g_m}$

wo: $\sqrt{2} \sqrt{\frac{g_m}{2 C_1 C_5 R_1 R_5 g_m + C_1 C_5 R_5}}$

bandwidth: $\frac{\sqrt{2} \sqrt{\frac{g_m}{2 C_1 C_5 R_1 R_5 g_m + C_1 C_5 R_5}} \left(2 C_1 R_1 g_m + C_1 + 2 C_5 R_5 g_m\right)}{2 \sqrt{2} \sqrt{C_1} \sqrt{C_5} R_1 \sqrt{R_5} g_m \sqrt{\frac{g_m}{2 R_1 g_m + 1}} + \sqrt{2} \sqrt{C_1} \sqrt{C_5} \sqrt{R_5} \sqrt{\frac{g_m}{2 R_1 g_m + 1}}}$

K-LP: $\frac{R_5 g_m - 1}{2 g_m}$

K-HP: $- \frac{R_1}{2 R_1 g_m + 1}$

K-BP: $\frac{C_1 R_1 R_5 g_m - C_1 R_1 - C_5 R_5}{2 C_1 R_1 g_m + C_1 + 2 C_5 R_5 g_m}$

Qz: None

Wz: $\frac{\sqrt{- R_5 g_m + 1}}{\sqrt{C_1} \sqrt{C_5} \sqrt{R_1} \sqrt{R_5}}$

11 X-PolynomialError